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MTCLF: A multitask curriculum learning framework for unbiased glaucoma screenings

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ABSTRACT

Background and Objective: Glaucoma is a disease that causes irreversible damage to the optic nerve. Research on accurate automatic screening algorithms is essential for the prevention and treatment of glaucoma. However, due to the imbalance of existing datasets and the existence of some hard samples that accompany other diverse and complex fundus diseases, the performance of current glaucoma screening algorithms is limited. In addition, the lack of interpretability also makes it difficult for the current algorithms to meet the requirements of clinical applications.
Method: In this paper, we propose a new multitask curriculum learning framework (MTCLF) for unbiased glaucoma screenings and visualizations of model decision-making areas. MTCLF is a teacher-student framework. The teacher network is used to generate the label evidence map. The student network can diagnose glaucoma and predict the evidence map at the same time with the well-designed dual-branch CNN structure and collaborative learning module. We design two curriculum coefficients θ and σ to guide the training process of the student network in the sample space so that the student network can adaptively balance the sample contribution, reduce the prediction bias and mine hard samples.
Results: The experimental results show that the accuracy, sensitivity, specificity, AUC and F_2-score of MTCLF based on the LAG dataset for glaucoma diagnoses are 0.967, 0.961, 0.970, 0.996, and 0.958, respectively. These results are superior to those of the state-of-the-art methods.
Conclusion: MTCLF not only achieves the best performance for unbiased glaucoma diagnoses but also generates a reliable evidence map to help clinicians explore fine lesion areas.

1. Introduction

Glaucoma has become the second leading eye disease that causes blindness in the world [1]. It threatens the visual health of more than 65 million people [2]. Glaucoma develops slowly, and its symptoms are mild in the early stages, making it easy for patients to overlook irreversible damage to their vision. Therefore, early screenings and treatment are essential for the prevention and treatment of glaucoma.

The common clinical diagnosis method for glaucoma is optic nerve head (ONH) examination, which refers to the comprehensive analysis of pathological phenomena and the physiological structures of glaucoma in fundus images by ophthalmologists. The main pathological changes are neural retinal edge erosion [3,4], optic cup dilation, defects on the retinal nerve fibre layer [5], optic disc haemorrhaging and β-parapapillary atrophy (β-PPA) [6]. These subtle lesions provide doctors with diagnostic evidence.

In this paper, we propose a new multitask curriculum learning framework (MTCLF) for the unbiased screening of glaucoma and the visualization of model decision-making areas. MTCLF is a teacher-student framework that achieves two main tasks: glaucoma diagnoses and evidence map predictions. The well-designed teacher convolutional neural network (CNN) is used to generate the label evidence map and provide a prior curriculum to assist the training of the student network. The student network uses a dual-branch CNN to diagnose glaucoma and predict the evidence map. The output of the evidence map prediction branch is also used to enhance the feature map of the glaucoma diagnosis branch. The training process of the student network is guided by its self-feedback curriculum and the prior curriculum provided by the teacher network. The loss of training samples is dynamically weighted to adaptively adjust the sample contribution so that it can overcome the prediction bias and mine hard samples. The main contributions of this article are as follows: i) MTCLF combines curriculum learning with multitasking methods to simultaneously satisfy the unbiased diagnosis of glaucoma and the prediction of evidence maps. It can help clinical ophthalmologists find subtle lesions in fundus images and assist doctors in the diagno-

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sis of glaucoma. ii) Two curriculum coefficients for the weighting loss are designed. They are the sample prior curriculum weighting coefficient θ provided by the teacher network and the sample self-feedback curriculum weighting coefficient σ provided by the student network. θ and σ are used to guide the training process of the student network to fully mine hard samples and overcome the problem of sample distribution imbalance and small differences among classes. iii) The well-designed teacher-student network ensures that MTCLF can fully improve the performance of unbiased glaucoma diagnoses and evidence map predictions. With the evidence map as a bridge, the sample prior information learned by the teacher network is transmitted to the student network so that the student network can enhance the feature map of its glaucoma diagnosis branch while realizing the prediction of the evidence map.

2. Related Works

In recent years, with the development of computer technology, many computer-aided diagnosis algorithms for glaucoma based on digital retinal images have been proposed. These automatic glaucoma diagnosis methods mainly include machine learning methods based on feature engineering and deep learning methods based on CNNs. Feature engineering-based methods rely on manually making and extracting some discriminative features of glaucoma from fundus images and training machine learning classifiers. The extracted features mainly include structural features with prior clinical information, such as the optic cup-to-disc ratio (CDR) [7], diameter of the optic disc [4], area of the optic cup, and ISNT [8] rule, or some image features, such as texture [9], wavelet-based [10,11], and Gabor-based [12] features. Manually made features are difficult to use when fully characterizing glaucoma, so the accuracy of glaucoma diagnosis by this method is limited. Deep learning methods integrate feature extraction and classification and have shown excellent performance in biomedical image analysis. Therefore, many new CNNs [13–17] have been designed to improve the diagnostic performance of glaucoma screenings. For example, Chai et al. [18] designed a multi-branch neural network (MB-NN) combining domain knowledge. The inputs of the three branches of MB-NN are the original image, the optic disc region extracted by the object detection algorithm, and the one-dimensional features containing domain knowledge such as age, intraocular pressure and eye sight. Zhao et al. [19] proposed MFPPNet to directly estimate CDR from fundus images to diagnose glaucoma by eliminating the intermediate step of segmenting OD and OC. MFPPNet uses three DenseBlocks to extract features from fundus images, and then learn to fuse multi-scale features through feature pyramid pooling module and fully connected feature fusion module. Finally, CDR regression is performed using random forest regression. Besides, reference [20] reviews recent advances in deep learning on fundus images, providing summaries and analyses for each task, including state-of-the-art methods for glaucoma screening. Although CNN models have made a certain breakthrough, they also have some widely criticized shortcomings, such as the lack of interpretability, which limits their clinical application.

Regardless of the automatic glaucoma diagnosis algorithm, performance bottlenecks will be encountered because of the following difficult challenges: **i) unbalanced training sample ratio**. Most open source datasets or clinical collection datasets have a large proportion of normal fundus images, while glaucoma-positive samples are generally rare. If a model is trained on a dataset with unbalanced category proportions, it will produce biased predictions; **ii) the existence of some hard samples affects the accuracy of a model**. In the early stage of glaucoma, the symptoms are mild and difficult to easily distinguish. Some patients have congenital optic nerve dysplasia or have other ophthalmological diseases and trauma. It is difficult for a model to correctly identify these rare hard samples, thereby reducing its sensitivity and accuracy, which is unacceptable in clinical applications. Fig. 1 shows some hard samples and non-hard samples; **iii) the differences between classes are small, but the differences within classes are large**. There are diversified types of glaucoma, and their characteristics in fundus images are different. The overall difference between glaucoma and nonglaucoma images is small, so the accuracy of a model is difficult to further improve; **iv) lack of interpretability**. Models such as CNN cannot provide diagnostic evidence due to their black box characteristics, which makes them unable to meet the standards of clinical applications.

Curriculum learning [21] provides new ideas for improving nonconvex optimization and model generalization performance. Classical curriculum learning sorts samples according to the degree of difficulty. During the training process, the training dataset is gradually expanded from simple samples to difficult samples to improve the performance of the trained model. Many studies have reported that curriculum learning can play a positive role in medical image processing problems, such as disease detection and classification [22–25] or lesion localization [26]. Although curriculum learning can enhance the generalization ability by training the model sequentially from difficult to easy, the existing methods still cannot completely overcome the abovementioned problems in the automatic diagnosis of glaucoma. In the cases of uneven distributions of training samples, the existence of difficult examples, and large interclass similarity, the training effect of a model cannot be further improved. More importantly, these methods often lack interpretability and cannot visualize decision-making evidence for the diagnosis of glaucoma.

Therefore, it is important to propose an unbiased and interpretable glaucoma screening algorithm to solve the shortcomings of existing algorithms. The unbiased glaucoma screening means that the algorithm can effectively overcome the training bias introduced by the imbalance of the training data and reduce false negatives in predicted results. In addition, such algorithms can overcome the adverse effects caused by the hard samples in the training data and reduce false positive rate (FPR) and false negative rate (FNR) of the screening results, that is, improve specificity and sensitivity.

3. Proposed Method

The proposed MTCLF method is shown in Fig. 2. It consists of a well-designed student network, teacher network and curriculum module. The teacher network is designed to initially screen for glaucoma and obtain a label evidence map that reflects sample information and the basis for diagnoses. The student network is a multitask network that simultaneously predicts the evidence map and completes an unbiased glaucoma screening. The curriculum module is generated to ensure that the student network conducts an unbiased glaucoma screening. It designs two curriculum weighting coefficients θ and σ by making full use of the sample prior information provided by the trained teacher network and the feedback information of the student network training process to dynamically adjust the training loss of the student network, so that the student network can balance the sample distribution and overcome the interference of hard samples.

3.1. Teacher network based on a self-attention module

In the proposed MTCLF, the role of the teacher network is to provide prior information to assist the training of the student network. On the one hand, the teacher network generates activation maps with key regions features for glaucoma classification as the label evidence maps for the student network, on the other hand,

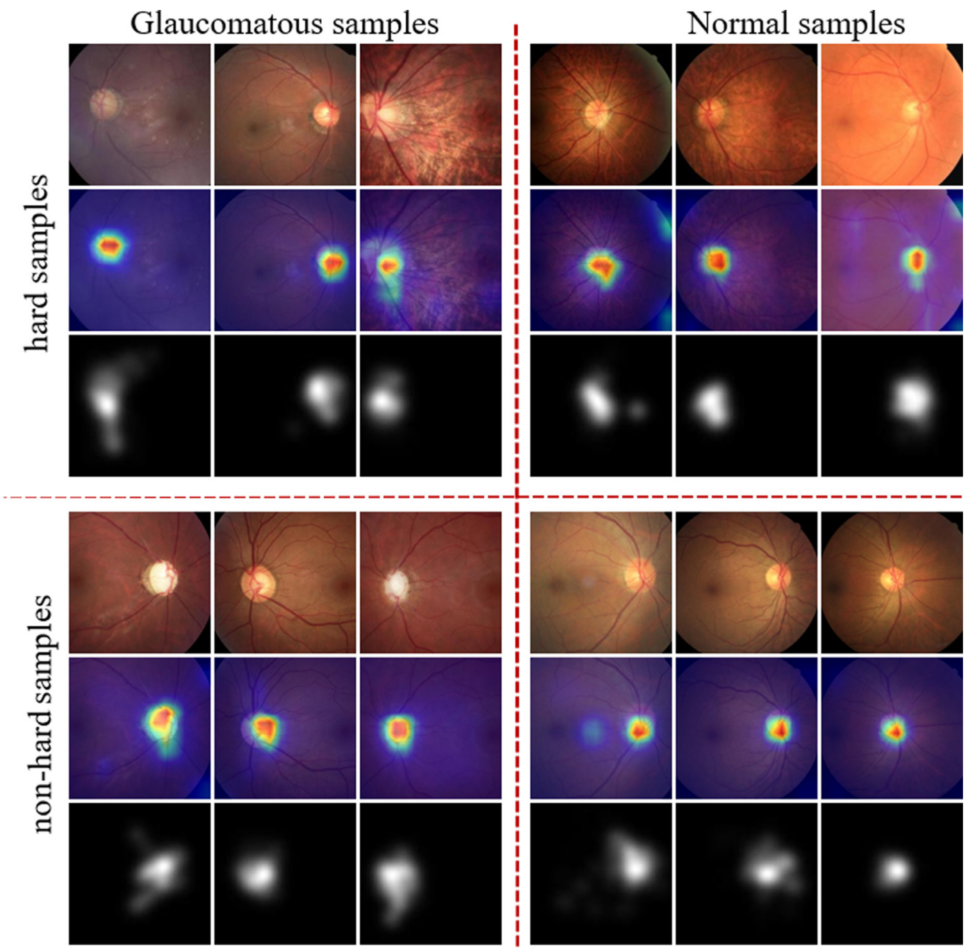

Fig. 1. Some hard samples and non-hard samples. The first row shows the fundus image, the second row shows the evidence maps, and the third row shows the attention areas marked by doctors [15].

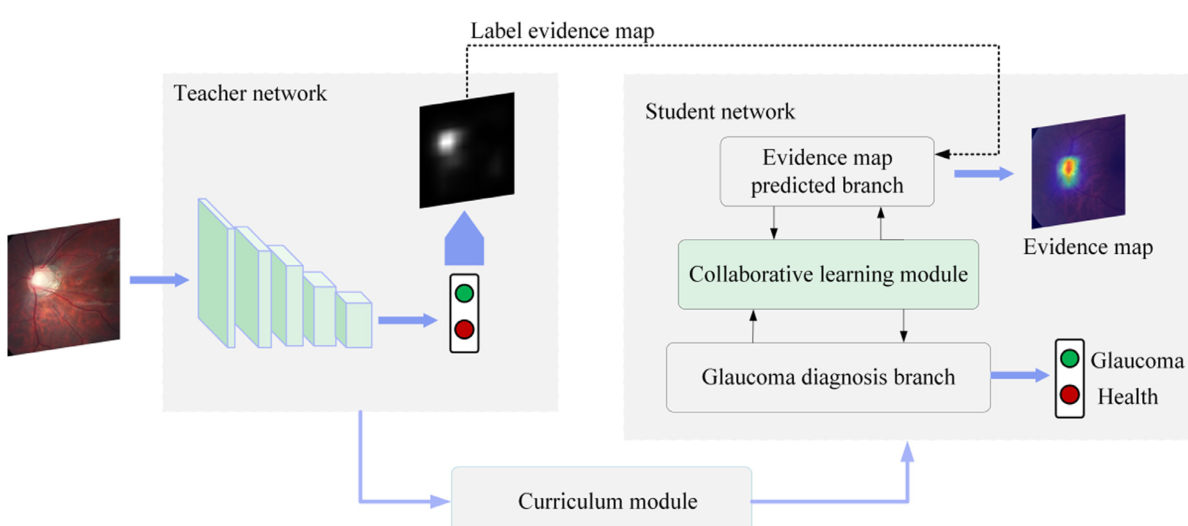

Fig. 2. The proposed MTCLF model for glaucoma screening.

its glaucoma classification results are used to generate curriculums to guide the training of the student network.

To ensure the quality of the generated evidence map and provide reliable preliminary glaucoma classification results, the designed teacher network uses ResNet-34 [27] as the backbone and extracts semantic feature maps of different depths to construct the multi-scale discrimination module, and adopts the self-attention mechanism module to make the network pay attention to spatial information and channel information at the same time. Its network structure is shown in Fig. 3.

Specifically, ResNet-34 consists of a series of residual blocks and pooling layers. We named the sets of feature maps before each pooling layer the 1st to 5th sets of feature maps. In this method, we downsample the 2nd to 5th sets of feature maps to a size of

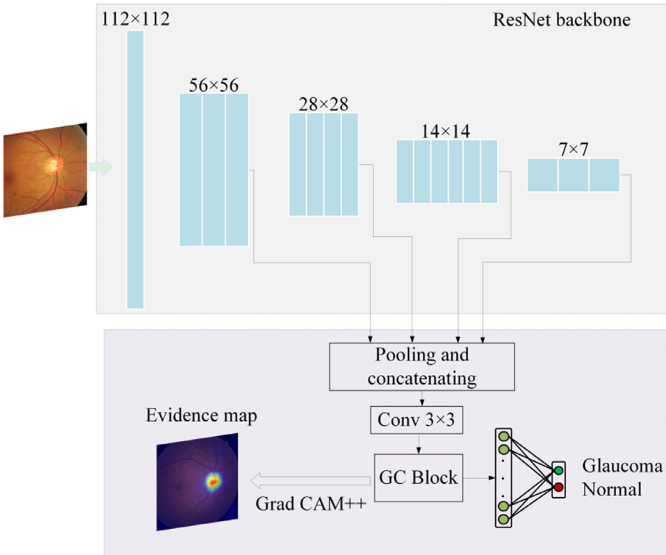

Fig. 3. The architecture of the teacher network.

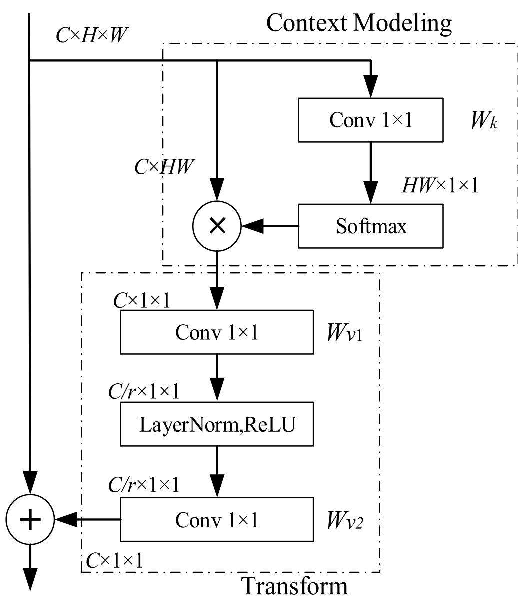

Fig. 4. The architecture of the GC block.

7 × 7 and concatenate them together. The channel dimensionality of the concatenated feature maps is reduced through a 3 × 3 convolution, and then the global context information of the features and the correlation among the channels of different feature maps are constructed with the Global Context (GC) block [28]. The feature maps obtained by the GC block are subjected to a global mean pooling operation and then concatenated to form a fully connected layer as a glaucoma classifier.

The GC block is a new global context modelling framework and is a simplified version of nonlocal block [29]. It can not only effectively model long-range dependencies but also has the advantage of the squeeze-excitation (SE) block [30], which is the channel attention mechanism. The architecture of the GC block is shown in Fig. 4 and is formulated as

$$\mathbf{z}_i = \mathbf{x}_i \oplus W_{v2}\mathrm{ReLU}\left[\mathrm{LN}\left(W_{v1}\sum_{j=1}^{N}\frac{e^{W_k \mathbf{x}_j}}{\sum_{m=1}^{N} e^{W_k \mathbf{x}_m}}\mathbf{x}_j\right)\right] \quad (1)$$

where $\mathbf{x}$ represents the input feature maps, $\mathbf{z}$ is the output feature maps, and $\oplus$ refers to pixel-level addition. N represents the number of feature maps, W refers to the parameter of the convolution operation, and $e^{W_k \mathbf{x}_j} / \sum_{m=1}^{N} e^{W_k \mathbf{x}_m}$ can be regarded as a weight for global attention pooling. LN is the layer normalization used to simplify optimization and serves as a regularizer.

After teacher network training is completed, the evidence map is generated through the Grad CAM++ [31] algorithm and provided to the student network. The evidence map can reflect the key discriminative areas of the input fundus image, highlighting local spatial features. It plays two important roles. First, it serves as the label of the evidence map predicted branch of the student network. Second, it provides prior knowledge for the training of the student network and reweights the deep feature maps of the student network to enhance the ability to detect difficult samples.

The teacher network reorganizes feature maps of different depths and then uses a self-attention mechanism to construct a classifier for glaucoma, which can improve the reuse rate of features and enhance the network's ability to extract discriminative features. In addition, it not only helps to discover the evidence map of key features of the space but also provides guarantees for the generation of the curriculum.

3.2. Multitask student network

The student network is a multitask CNN composed of two interrelated branches with different lengths. The short branch predicts the evidence map of the input fundus image, and the long-branch performs an accurate diagnosis of glaucoma. The evidence map is used to guide the construction of long branches. Specifically, it is used to weight the feature maps of the glaucoma prediction branches to strengthen the relevant areas for the glaucoma screening in the feature maps and suppress nonrelevant areas. Therefore, the extraction effect of the network on the key disease features of glaucoma can be significantly enhanced. In addition, the designed collaborative learning module (CLM) is used between the two branches to realize feature exchange and enhancement of the two branches.

The student network is trained according to the combined loss function of the designed evidence map prediction loss and glaucoma classification loss and is guided by the generated curriculum during the training process, and the loss function is dynamically adjusted in the sample space. Therefore, the student network can overcome the interference of difficult samples while accelerating the training and complete dual tasks accurately at the same time in a single network. The structure of the student network is shown in Fig. 5.

1) Dual-branch student network

The student network uses ResBlock [27] as the basic feature extraction module. The glaucoma prediction branch consists of five-stage convolution modules. Each stage includes several ResBlocks and is followed by a pooling operation to reduce the size of the feature maps. As shown in Fig. 5, we symbolize the output feature maps of all stages as $\mathbf{G}_1$ to $\mathbf{G}_5$ in order. The evidence map prediction branch consists of three-stage convolution modules, which are the same as the first three stages of the glaucoma prediction branch. Their output feature maps are denoted as $\mathbf{E}_1$, $\mathbf{E}_2$, and $\mathbf{E}_3$, respectively. To predict the evidence map more accurately and highlight the critical areas for glaucoma diagnoses in the fundus images, the output feature maps of the three stages are mapped to the same spatial size as $\mathbf{E}_2$ and then decoded by the ASPP [32] module to construct the evidence map (EM).

EM is not only used as the output of the evidence map prediction branch but also used for broadcast multiplication with $\mathbf{G}_3$ of

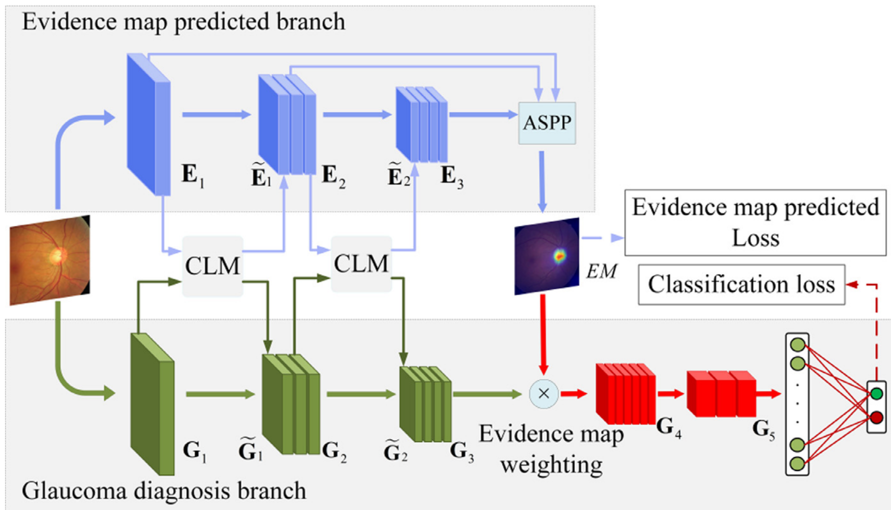

Fig. 5. The structure of the student network.

the glaucoma prediction branch, which allows the glaucoma prediction branch to focus on the key distinguishing features based on the prior information provided by the evidence map. Thus, the performance of the glaucoma prediction can be greatly improved.

In a serialized CNN composed of basic computing modules, to increase the receptive field of the convolution operation and avoid multiple increases in the calculation amount, pooling operations are used to reduce the scale of the feature map after each calculation stage. Although pooling can achieve translation invariance, it also leads to the loss of the precise spatial relationships among local objects in the image. In terms of the clinical diagnosis of glaucoma, the relative spatial position and size of the optic cup, optic disc and optic nerve rim highlight many important indicators, such as CDR, that are very important for the diagnosis of glaucoma. In addition, other subtle lesion areas related to glaucoma in the fundus image, such as optic disc haemorrhaging, β-PPA, and optic nerve fibre layer damage, are easily lost after multiple pooling operations. EM generated by the teacher network carries this key spatial relative position information. Therefore, the student network uses EM to construct an attention mechanism to integrate the target edge features and spatial information with $\mathbf{G}_3$ to strengthen the relevant areas of a glaucoma screening and suppress the nonrelevant areas.

2) Collaborative learning module

The student network can simultaneously implement the semantic segmentation task of evidence map predictions and the binary classification task of positive and negative samples for glaucoma screenings. Although the output results of these two tasks have different modalities, they both rely on the semantic features extracted by the CNN that reflect the size, location, edge, and texture of the subtle lesions. These features have a strong commonality. To enable the student network to achieve the expected results in dual-task predictions, we adopted the designed CLM between the two branches. As a bridge between the two branches, the CLM can play a role in sharing interactive features and can effectively improve the accuracy of glaucoma screenings and the accuracy of evidence map predictions.

The details of the CLM are shown in Fig. 6. Given $\mathbf{G}_i$ of the glaucoma screening branch and $\mathbf{E}_i$ of the evidence map prediction branch, we concatenate $\mathbf{G}_i$ and $\mathbf{E}_i$ and use convolution operations to learn the mapping relationships of feature interactions specific to different tasks. Then, the learned interactive feature maps are merged with the original feature maps through pixel-level addition so that the interactive features can share and exchange information with the original features by learning residuals.

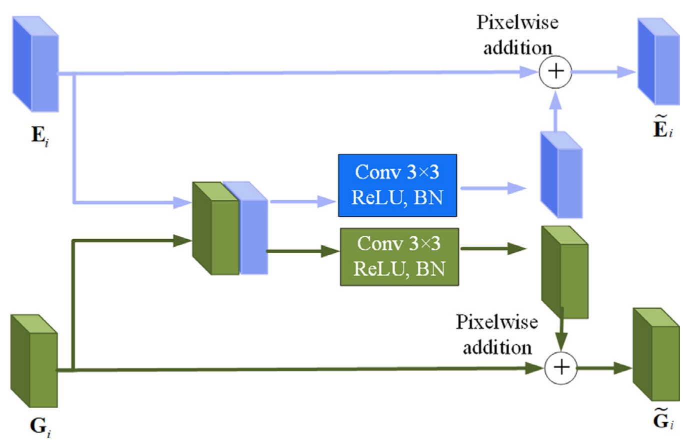

Fig. 6. The network architecture of the CLM.

$\tilde{\mathbf{G}}_i$ and $\tilde{\mathbf{E}}_i$ represent the feature maps of $\mathbf{G}_i$ and $\mathbf{E}_i$ after the feature interaction operations, respectively, and they are formulated as

$$\tilde{\mathbf{G}}_i = \mathrm{g}[\mathbf{G}_i, \mathbf{E}_i] \oplus \mathbf{G}_i \tag{2}$$

$$\tilde{\mathbf{E}}_i = \mathrm{f}[\mathbf{G}_i, \mathbf{E}_i] \oplus \mathbf{E}_i \tag{3}$$

where [,] represents the concatenation of the feature maps and g and f represent the 3 × 3 convolutions used to learn the feature interaction mapping relationship.

3) Multitask loss function

We designed a multitask loss function that includes the classification loss and evidence map prediction loss for dual tasks to achieve collaborative learning. The classification loss L^{class} uses the cross entropy loss, which is formulated as follows:

$$L^{\text{class}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y^i \cdot \log p^i + \left(1 - y^i\right) \cdot \log\left(1 - p^i\right)\right] \tag{4}$$

where p^i refers to the probability that sample i is predicted as the target label, and y^i represents the ground truth label.

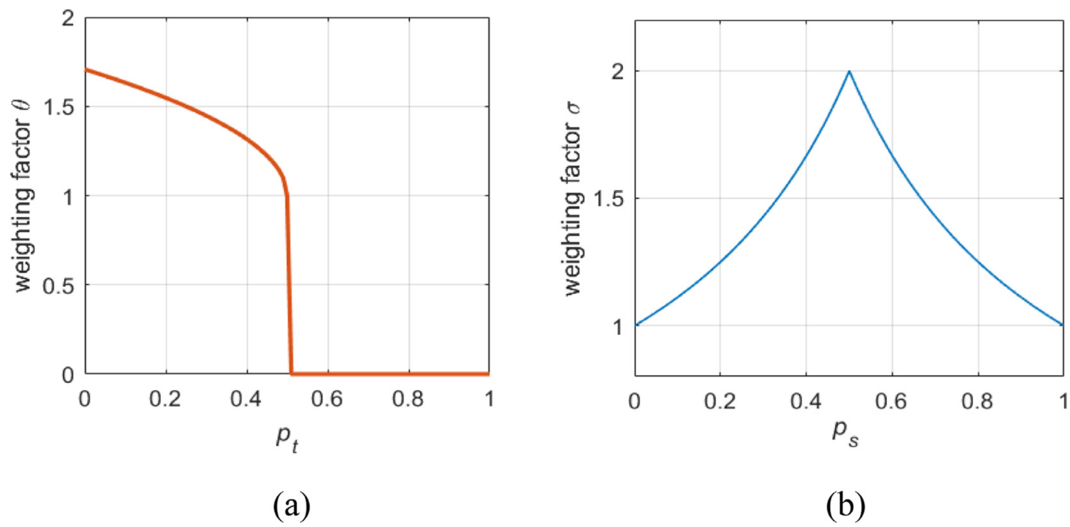

Fig. 7. The dynamic change in the weighting factors θ and σ.

For evidence map prediction, we consider it as a pixel-level binary classification task and adopt binary cross-entropy loss as its loss function L^{EM}, whose formula is the same as L^{class}. Hence, the loss function of the dual-task student network is:

$$L^{\text{total}} = L^{EM} + L^{\text{class}} \tag{5}$$

3.3. Curriculum module

The uneven distribution of negative and positive samples in the existing glaucoma database can easily affect the training process of the model and makes the trained model seriously biased toward the negative samples, resulting in lower sensitivity. In addition, the CNN model has poor learning ability for rare hard samples, which also affects the performance of glaucoma screenings.

To solve the above two problems, we designed a curriculum-based learning strategy in the sample space specifically for the student network. By making full use of the sample prior information provided by the trained teacher network and the feedback information during student network training, we designed two coefficients θ and σ to dynamically weight the loss of each sample in each batch in the training process. Thus, the student network can focus on difficult samples and improve the classification ability of hard samples. In addition, the designed curriculum can adaptively adjust the contribution of training samples to balance the distribution of positive and negative samples. Specifically, the loss function of the student network guided by the curriculum is:

$$L^{\text{total}} = L^{EM} + [1 + \lambda\theta + (1-\lambda)\sigma]L^{\text{class}} \tag{6}$$

where λ refers to the equalization coefficient, which is set to 0.5. θ is the sample prior loss coefficient, which reflects the prior information provided by the teacher network about positive samples that are difficult to classify. With the guidance of the trained teacher network, the student network can accelerate the learning of hard glaucoma-positive samples. In each training iteration, the sample is input into the trained teacher network to calculate the confidence p_t, and θ is calculated by the following formula:

$$\theta^i = \mu_t^i \left(\left| p_t^i - 0.5 \right|^{1/2} + 1 \right)$$

$$\mu_t^i = \begin{cases} 0 & y'_i == y_i \\ 1 & (y'_i \neq y_i) \text{ and } (y_i == 1) \end{cases} \tag{7}$$

where i refers to the sample number, and confidence p_t^i refers to the output value corresponding to the glaucoma category of the teacher network after softmax activation. y'_i represents the predicted label of the student network for sample i, and y_i is the ground truth label. μ_t^i is used to limit the range in which the curriculum coefficient θ works; only when the student network predicts the glaucoma-positive samples incorrectly does θ play a role in the weighted adjustment of the loss.

σ is the sample feedback loss coefficient and reflects the learning state of the student network during the training process, especially the recognition ability for hard samples. By reweighting the loss of misclassified samples, the student network can adjust itself to mine hard samples and balance training gains. σ is calculated by the following formula:

$$\sigma^i = \frac{\mu_s^i}{\left(\left| p_s^i - 0.5 \right| + 0.5 \right)}$$

$$\mu_s^i = \begin{cases} 0 & y'_i == y_i \\ 1 & y'_i \neq y_i \end{cases} \tag{8}$$

where p_s^i refers to the output value corresponding to the glaucoma category of the student network after softmax activation. μ_s^i ensures that σ^i only works when the student network predicts both positive and negative samples incorrectly. Figs. 7(a) and 7(b) show the dynamic changes in the weighting factors θ and σ, respectively.

Fig. 7(a) shows that when the glaucoma-positive training sample i is tested by the teacher network with confidence $p_t^i < 0.5$ and is misclassified, it is recognized as a hard sample. Therefore, the weighting factor θ increases the weight of L^{clsss} of sample i, but it is invalid for other samples. It is very important that the closer p_t^i is to 0, the greater the degree of misclassification of sample i, so θ is increased to promote the student network to pay more attention to sample i.

In addition, Fig. 7(b) clearly shows that σ focuses on the misclassified samples of all categories and especially focuses on the ambiguous misclassified samples with a high degree of similarity between classes. In the training process of the student network, when sample i is misclassified by the student network and its glaucoma category confidence p_s^i is close to 0.5, it is in the critical region of classification. In this case, σ increases and makes the student network pay more attention to the critical samples so that the training process can be accelerated and ambiguous samples with high similarity among classes can be mined.

4. Experiments and Results

4.1. Datasets

Four datasets are used to evaluate the proposed MTCLF model for glaucoma screenings: LAG [15], ORIGA [33], REFUGE [34], and RIM-ONE-r2 [35].

The LAG database is the largest fundus image dataset that contains glaucoma expert labels and was published recently. The data set contains 4,854 colour fundus images, including 1,711 fundus images of suspected glaucoma patients and 3,143 nonglaucoma fundus images. All fundus images in the dataset were diagnosed by qualified glaucoma experts who comprehensively considered the intraocular pressure, visual field, and optic disc disease. In our experiment, the dataset is randomly divided into a training set and a test set at a ratio of 8:2. For the training set we use a stratified 5-fold cross validation method to further divide it into training set and validation set, and we compute the average number of training epochs for cross-validation as the hyperparameter epoch when retraining. Finally the experiments are re-run on the full training set to test the advantages of the method.

To further verify the robustness of MTCLF and to compare it fairly with other state-of-the-art methods, we chose the other three most commonly used datasets, namely, ORIGA, REFUGE and RIM-ONE-r2. ORIGA is composed of 650 fundus images and their glaucoma diagnosis results, of which 168 are glaucomatous and 482 are normal. The REFUGE challenge publicly released a dataset of 1,200 fundus images with clinical glaucoma diagnostic labels, of which 120 are glaucomatous and 1,080 are non-glaucoma. RIM-ONE-r2 consists of 200 glaucomatous and 255 normal fundus images. The positive and negative sample distributions of ORIGA and REFUGE are seriously unbalanced, while that of RIM-ONE-r2 is approximately balanced. Therefore, these datasets are specifically used to compare the glaucoma diagnostic performance of different algorithms under unbalanced and balanced sample distributions. The training set and test set division ratio of these datasets are consistent with the LAG dataset. For both the teacher network and the student network, all input images are resized to 224 × 224.

4.2. Implementation details

In our proposed MTCLF method, the training process is implemented in two stages. In the first stage, the input image and corresponding diagnosis labels are used to supervise the training of the teacher network to obtain the optimal teacher network model and generate the evidence maps. In the second stage, in addition to the glaucoma diagnosis label, the student network also uses the evidence map generated by the teacher network as the target of its evidence map prediction branch and obtains prior information on the semantic features. During the training process, the loss function is dynamically adjusted according to the designed curriculum weighting coefficients θ and σ. Both stages of training use the Adam optimizer and cosine annealing learning rate adjustment strategy.

For the testing phase, by inputting fundus images to the trained student network, the network can simultaneously diagnose glaucoma, infer the decision-making basis of the network and help doctors diagnose glaucoma. The experiments are implemented based on the PyTorch platform, Intel Xeon E5-2678 v3 CPU and a GeForce RTX 2080 Ti graphics card (GPU).

4.3. Evaluation metrics

To fully evaluate the performance of the proposed method, we used a variety of different evaluation metrics. Specifically, the accuracy, sensitivity, and specificity are formulated as follows.

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{9}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{10}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN}+\text{FP}} \tag{11}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

In addition, the area under the receiver operating characteristic curve (AUC) and F_2-score were used to evaluate the performance of glaucoma screenings. F_2-score is the weighted harmonic mean of Recall and Precision. Recall, Precision and F_2-score are defined as follows. The reason for choosing F_2-score is that Recall is more important than the Precision under its calculation standard, and it is more suitable for the evaluation of glaucoma screening models. Since the glaucoma datasets are unevenly distributed and the proportion of positive samples is too small, Recall can better reflect the screening effect of positive samples, especially hard glaucoma samples. Besides, other papers on glaucoma screening [15,25,36,37] also use F_2-score as an evaluation metric, so we choose F_2-score for fair comparison with the data in these papers.

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{13}$$

$$\text{F}_2\text{-score} = \frac{5}{\frac{1}{\text{Precision}}+\frac{4}{\text{Recall}}} = \frac{5\text{TP}}{5\text{TP}+4\text{FN}+\text{FP}} \tag{14}$$

4.4. Glaucoma classification evaluation

Performance Comparison

In this section, to evaluate the glaucoma screening performance of the proposed MTCLF model, we compare several state-of-the-art glaucoma screening methods based on the LAG dataset. The candidate methods are all deep learning methods, including large-scale CNN methods, such as AG-CNN [15] and DENet [16], that integrate multiple tasks and small-scale CNN methods, such as Chen et al. [13] and EAMNet [17]. ResNet-50 [27] is also used as a baseline to compare the performance of MTCLF. These methods have been reported to have excellent performance in the automatic diagnosis of glaucoma. Specifically, AG-CNN [15] includes the attention prediction subnet, the pathological area location subnet, and the glaucoma classification subnet. The first two subnets complete specific tasks to serve the glaucoma classification subnet. DENet [16] integrates 4 detection branches to independently perform glaucoma screenings from the global fundus image, optic disc region of interest, polar coordinate transformed optic disc, and segmentation guidance features and finally votes to determine the screening result. Chen et al. [13] proposes a 6-layer CNN structure and uses contextualized training. EAMNet [17] applies multilayer average pooling (MLAP) to integrate the features of different layers of ResNet for glaucoma diagnoses.

In our method, the average training epoch of stratified 5-fold cross-validation is 56, and we retrain the model on the entire training set based on this determined epoch, and perform glaucoma diagnostic evaluation on the test set. The experimental comparison results are shown in Table 1, some of the experimental data are cited from [15,25,36,37]. It can be seen from the experimental results that although the proposed MTCLF model uses ResBlock as the basic computing unit, compared with ResNet-50, this

Table 1
Evaluation results of glaucoma diagnostic on the LAG dataset.

Method	Accuracy	Sensitivity	Specificity	AUC	F_2-score
ResNet [27]	0.938	0.910	0.954	0.978	0.931
DENet [16]	0.756	0.631	0.843	0.822	0.650
AG-CNN [15]	0.953	0.954	0.952	0.975	0.951
EAMNet [17]	0.934	0.927	0.938	0.978	0.932
Chen et al. [13]	0.892	0.906	0.882	0.956	0.894
Proposed MTCLF	0.967	0.961	0.970	0.996	0.958

method is superior with respect to many indicators and obtains improvements of 3.1%, 5.6%, 1.8%, and 2.9% in terms of accuracy, sensitivity, AUC and F_2-score. Compared with other state-of-the-art methods, the proposed MTCLF model achieves the best performance in the diagnosis of glaucoma, with the highest accuracy, sensitivity, AUC and F_2-score indicator values. Although the specificity score of MTCLF is not optimal, its sensitivity score is 52.3%, 0.7%, 3.7% and 6.1% higher than those of DENet, AG-CNN, EAMNet and Chen et al., respectively, which shows that the designed curriculum for hard sample mining and data balancing enables MTCLF to achieve the best sensitivity while maintaining excellent specificity. In addition, the F_2-score of our method is 47.4%, 0.7%, 2.8% and 7.2% higher than those of DENet, AG-CNN, EAMNet and Chen et al., respectively, which also confirms that this method can best balance sensitivity and specificity and can achieve the best glaucoma screening performance.

To further test the performance advantages of the proposed method, experiments were carried out based on three other high-quality datasets, namely, REFUGE, ORIGA and RIM-ONE-r2. Here we mainly compare typical deep learning methods, such as EAMNet, Chen et al. and some feature engineering-based methods such as M-Net [38] + CDR estimation, Wavelet [11], GLCM [9]. The M-Net + CDR estimation method is completely based on prior clinical information and can best reflect the level of clinical diagnosis. It uses the CDR calculated from the segmentation results of the optic disc and optic cup as the distinguishing feature to screen for glaucoma. M-Net is a U-shaped CNN composed of a multiscale input layer, a side output layer and a multilabel loss function. It can obtain reliable segmentation results of the optic cup and optic disc by applying a polar coordinate transformation. Wavelet and GLCM are classic feature engineering methods that rely on wavelet transform features and grey-level cooccurrence matrix texture features collected manually from fundus images that reflect the pathological changes in glaucoma, and finally train machine learning classifiers for glaucoma diagnoses.

The distributions of negative and positive samples in the ORIGA and REFUGE datasets are unbalanced. The ratio of negative and positive samples in ORIGA is 482:168, while that in REFUGE is 1080:120, which is extremely unbalanced. Table 2 shows the evaluation results of glaucoma diagnostic performance based on ORIGA dataset. It can be seen that the sensitivity scores of M-Net + CDR estimation, Wavelet, and GLCM methods are severely affected when faced with an imbalanced dataset, which weakens their ability to diagnose glaucoma. The glaucoma diagnostic performance of our proposed MTCLF is much better than that of other methods, with accuracy, sensitivity, specificity, AUC and F_2-score values of 0.897, 0.857, 0.909, 0.929 and 0.882, respectively. In terms of F_2-score, MTCLF scores 36.1%, 41.1%, 66.7%, 8.5% and 23% higher than M-Net + CDR estimation, Wavelet, GLCM, EAMNet and Chen et al., respectively.

Table 3 shows the evaluation results on REFUGE dataset. In the experiments, the SEDC [37] method is also used for the comparison of glaucoma diagnostic performance. SEDC is an adaptive rebalancing strategy in the feature space to improve the glaucoma diagnosis on imbalanced data by augmenting feature distribution with feature distilling and feature re-weighting. The results show that feature engineering-based methods cannot work at all when faced with datasets with severely imbalanced distributions, and their sensitivity is extremely low. While common deep learning methods such as EAMNet and Chen et al. do poorly in the balance of sensitivity and specificity, their F_2-scores are 0.731 and 0.644, respectively. The SEDC method with data equalization ability can achieve good results, with sensitivity, specificity and F_2-score of 0.800, 0.969 and 0.788, respectively. The proposed MTCLF achieves the best results on the REFUGE dataset, with sensitivity, specificity, and F_2-score of 0.875, 0.999, and 0.897, which are 9.4%, 3.1%, and 13.8% higher than SEDC, respectively.

The sample distribution of the RIM-ONE-r2 dataset was relatively balanced, and the evaluation results of glaucoma diagnostic performance based on it are shown in Table 4. When trained on the RIM-ONE-r2 dataset, M-Net + CDR estimation shows a good balance between sensitivity and specificity. However, EAMNet, Chen et al., Wavelet and GLCM still have poor sensitivity, which shows that these methods are greatly affected by difficult samples and cannot accurately distinguish the difference between glaucoma and normal samples. Table 4 shows that MTCLF can also achieve the best glaucoma diagnostic performance on the RIM-ONE-r2 dataset, with accuracy, sensitivity, specificity, AUC and F_2-score values of 0.891, 0.850, 0.922, 0.924 and 0.885, respectively.

MTCLF relies on the prior knowledge provided by the designed curriculum weighting coefficient and evidence map to enable its student network to fully mine difficult samples during the training process, enhance the ability to recognize glaucoma-positive samples, and further reduce the false diagnosis rate. The superior performance on different datasets confirms that MTCLF can overcome the uneven distribution of samples and achieve high accuracy in glaucoma screenings while maintaining a balance between sensitivity and specificity. In addition, it also has reliable robustness and can be applied to different datasets.

Ablation Study

We conduct ablation experiments based on the LAG dataset to explore two questions: 1. the impact of the designed teacher-student two-stage method on the performance of glaucoma diagnosis, 2. the effects of the designed curriculum weighting coefficients θ and σ on hard sample mining. The LAG dataset is chosen because it is the largest fundus image dataset covering various glaucoma situations.

On the one hand, we take the method of using only the student network for glaucoma detection as a one-stage baseline, which does not have the EM and curriculum coefficients generated by the teacher network to guide its training. On the other hand, we use the teacher-student network without the guidance of the curriculum coefficient as the two-stage baseline and compare the performance of glaucoma diagnoses under the guidance of the single-curriculum coefficient and the dual-curriculum coefficient. The results of the ablation experiment are shown in Table 5 and Fig. 8 shows the ROC curves of the methods.

The one-stage baseline can achieve an accuracy of 0.940, which is almost the same as the two-stage baseline. However, the one-stage baseline is weaker in the recall of glaucoma-positive samples, specifically, the two-stage baseline is 2.5% and 4.2% higher than the one-stage baseline in sensitivity and F_2-score, respectively. The two-stage baseline uses the label EM provided by the teacher network as the target output of its evidence map prediction branch, and the EM output by the student network is also used as an attention mechanism for its deep feature map. In this way, the student network can additionally learn some key prior features that reflect pathological regions, thereby enhancing its ability to identify glaucoma-positive samples.

Table 2
Evaluation results of glaucoma diagnostic performance on the ORIGA dataset.

Method	Accuracy	Sensitivity	Specificity	AUC	F_2-score
EAMNet [17]	0.812	0.815	0.811	0.884	0.813
Chen et al. [13]	0.788	0.700	0.857	0.851	0.717
M-Net + CDR estimation [38]	0.749	0.621	0.878	0.818	0.648
Wavelet [9]	0.699	0.642	0.756	0.747	0.625
GLCM [11]	0.643	0.540	0.746	0.728	0.529
Proposed MTCLF	0.897	0.857	0.909	0.929	0.882

Table 3
Evaluation results of glaucoma diagnostic performance on the REFUGE dataset.

Method	Accuracy	Sensitivity	Specificity	AUC	F_2-score
EAMNet [17]	0.917	0.790	0.931	0.953	0.731
Chen et al. [13]	0.892	0.708	0.912	0.947	0.644
SEDC [37]	0.953	0.800	0.969	0.933	0.788
M-Net + CDR estimation [38]	0.766	0.583	0.949	0.819	0.579
Wavelet [9]	0.715	0.500	0.931	0.798	0.487
GLCM [11]	0.657	0.333	0.981	0.735	0.370
Proposed MTCLF	0.988	0.875	0.999	0.990	0.897

Table 4
Evaluation results of glaucoma performance on the RIM-ONE-R2 dataset.

Method	Accuracy	Sensitivity	Specificity	AUC	F_2-score
EAMNet [17]	0.832	0.800	0.818	0.857	0.803
Chen et al. [13]	0.800	0.696	0.870	0.831	0.711
M-Net + CDR estimation [38]	0.796	0.800	0.792	0.876	0.805
Wavelet [9]	0.707	0.650	0.765	0.792	0.657
GLCM [11]	0.772	0.700	0.843	0.888	0.714
Proposed MTCLF	0.891	0.850	0.922	0.924	0.885

Table 5
Ablation study based on the LAG dataset.

Method	Accuracy	Sensitivity	Specificity	AUC	F_2-score
one-stage baseline	0.940	0.883	0.971	0.983	0.895
two-stage baseline	0.942	0.905	0.964	0.977	0.933
two-stage baseline+weighted cross entropy	0.937	0.939	0.948	0.979	0.938
two-stage baseline+Focal loss [39]	0.948	0.947	0.949	0.990	0.948
two-stage baseline+θ	0.944	0.947	0.943	0.980	0.945
two-stage baseline+σ	0.949	0.942	0.952	0.987	0.947
two-stage baseline+θ+σ(MTCLF)	0.967	0.961	0.970	0.996	0.958

Although the F_2-score of the two-stage baseline is much better than that of the one-stage baseline and can achieve an excellent accuracy of 0.942, its specificity is higher than its sensitivity by 6.5%, which shows that the two-stage baseline is biased toward the negative sample category. The reason for the prediction bias is the unbalanced distribution of training samples for glaucoma-positive categories and the existence of a large number of hard samples. Compared with the two-stage baseline, the designed curriculum weighting coefficient improves the sensitivity significantly.

θ carries the prior information of the positive hard samples, which makes the student network focus on the training data identified as difficult samples by the teacher network and increases their training loss. Thus, when θ is adopted, the sensitivity of the two-stage baseline increases from 0.905 to 0.947. Although the specificity dropped by 2.2%, the F_2-score used to measure the overall diagnostic performance of glaucoma improved by 1.3%. σ allows the two-stage baseline to obtain feedback from the student network to focus on difficult-to-classify samples, especially misclassified samples with high similarity among classes. After the addition of σ, the sensitivity, AUC and F_2-score values of the two-stage baseline increased by 4.1%, 1.0% and 1.5%, respectively.

When the two curriculum coefficients are adopted, the glaucoma diagnosis performance is greatly improved. The best sensitivity and accuracy scores can be achieved while keeping the specificity stable. In addition, the AUC value increases from 0.977 to 0.996, and the F_2-score increases from 0.933 to 0.958.

We also conducted experiments based on the two-stage baselines with several other loss functions that can alleviate data imbalance to compare with the proposed curriculum weighting coefficients, such as weighted cross entropy loss and focal loss [39]. Weighted cross entropy loss assigns different weights to the loss of samples of different categories. In this paper, the weights of the glaucoma category and the non-glaucoma category are set to 0.75:0.25 according to the distribution of the training samples. The results are shown in Table 5. The weighted cross entropy loss magnifies the training loss of the glaucoma category samples, which can alleviate the low sensitivity caused by the unbalanced data distribution, so that the sensitivity and specificity are balanced. However, weighted cross entropy loss can't improve the overall accuracy in this task.

Focal loss uses the weight factor α to balance the loss of different categories to alleviate the imbalance of data. Besides, it uses a power function to reduce the loss of easy-to-classify samples, so that the model can focus more on difficult-to-classify samples during training. The formula of focal loss is as

$$\mathrm{FL}\left(p^{'}\right)=\begin{cases}-\alpha\left(1-p^{'}\right)^{\gamma}\log\left(p^{'}\right) & y=1\\ -(1-\alpha)p^{'\gamma}\log\left(1-p^{'}\right) & y=0\end{cases} \tag{15}$$

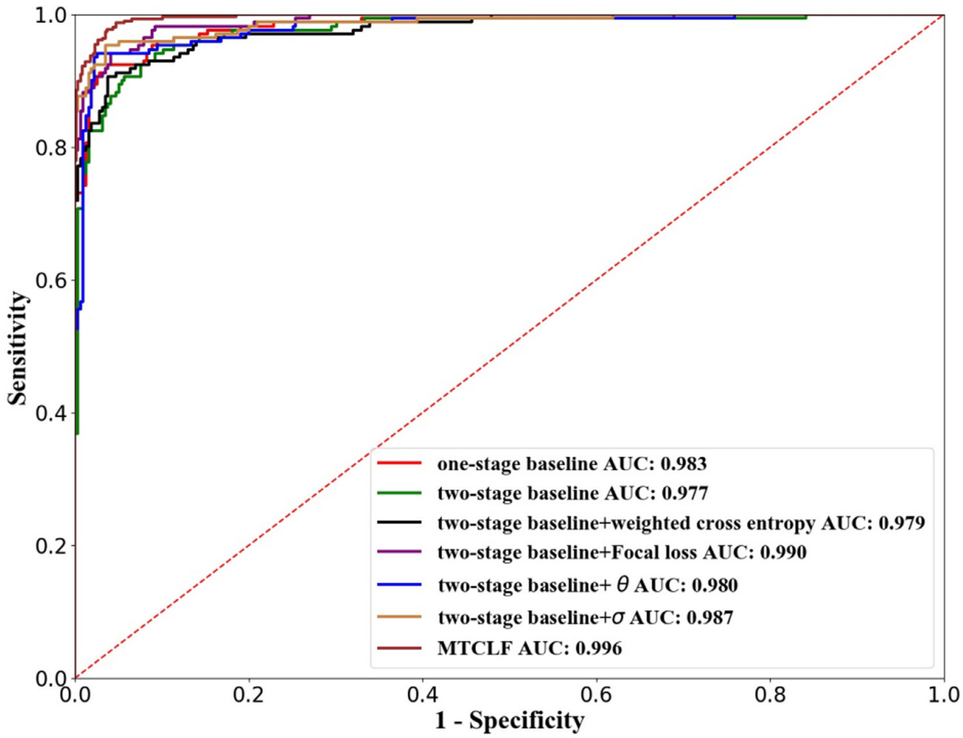

Fig. 8. ROC curves of different methods.

where p' refers to the probability that sample is predicted as the target label, and y represents the ground truth label. α is set to 0.25, γ is used to adjust the degree of weight reduction of easy sample loss, and is generally set to 2.

The experimental results show that focal loss can increase the sensitivity of the baseline by 4.6% and increase the overall accuracy by 0.6%. Through comparison, we can find that the proposed curriculum weighting coefficients can achieve better results than weighted cross-entropy loss and focus loss. The sensitivity of our method is 1.5% higher than that of focal loss, and the overall accuracy is also 2.0% higher than focal loss. The reason for achieving better results than focal loss is that focal loss will cause the model to pay too much attention to particularly difficult samples. Because the existence of $(1 - p')^{\gamma}$ in focal loss will greatly reduce the loss of well-classified samples, which is equivalent to extremely increase the weighted loss rate of difficult samples. This will cause some mislabeled samples or outliers to dominate the training process, which will adversely affect the training process. However, the proposed curriculum weighting coefficients limit the weighted multiple of the hard sample loss, as shown in Formula 7-8 and Fig. 7. Besides, the weighting coefficient θ carries a prior information, which provides more knowledge than focal loss. Therefore, the proposed method can achieve best results whether it is for the mining of hard samples or the balance of training loss deviation.

In short, the combined use of the curriculum weighting coefficients θ and σ enables the proposed MTCLF model to maximize its hard sample mining ability, overcome the interference of samples with high similarity among classes, and balance the contributions of training samples.

4.5. Evidence map generation evaluation

The evidence map output by the student network is a single-channel activation map with an intensity range of 0∼1, and the intensity of the region highly related to glaucoma detection is close to 1. To obtain the visualization result of the evidence map, the activation map is fused with the fundus image according to a certain proportion after pseudo-colour processing.

Given that the prediction of evidence maps is a pixel-level classification task, in this section, we use Accuracy and AUC to quantitatively evaluate the EM predicted by the student network based on the LAG test set. The results show that the EM predicted by MTCLF can achieve an accuracy of 0.820 and an AUC of 0.984 compared with the label evidence map.

To analyze the similarities and differences between the predicted EM and the label EM, we have carried out a large quantity of experiments. Fig. 9 and Fig. 10 show some illustrative examples. As can be seen in the figures, the student network can restore the label EM in detail, and the activation area of the predicted EM is consistent with the label EM, which is a small area centered on the optic disc, as shown in Fig. 9(c) and Fig. 10(c). The difference is that the key activation areas of the label EM are concentrated and rough, while the key activation areas of the predicted EM are more detailed and can provide more information after visualization. Although the output EM of the student network is trained with the label EM as the target, it is also an intermediate result in the student network, which is used as an attention map to weight the deep feature maps, so it can learn more specific information under the constraint of the label EM.

The most important factor in the clinical diagnosis of glaucoma is whether the optic nerve rim is losing area. If the optic nerve rim narrows slightly or significantly or if signs of optic nerve rim tangency appear, there is a high possibility of glaucoma. A large quantity of experiments was carried out with the help of professional doctors. Fig. 9 presents the fundus images of optic nerve rim loss in the LAG dataset and the corresponding visual evidence map generated by MTCLF.

Example #1 in Fig. 9 is a glaucoma fundus image with an obvious loss of the optic nerve rim. The optic nerve rims on the upper and lower sides of the optic disc are severely lost and show obvious signs of tangency. The evidence map generated by MTCLF

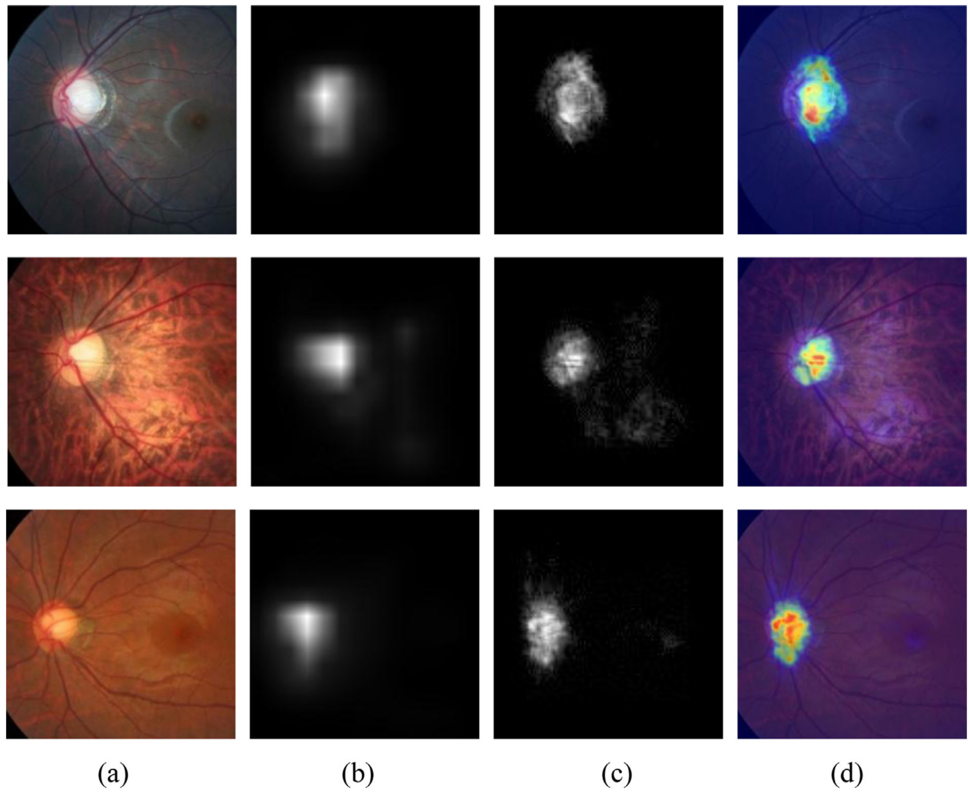

Fig. 9. Glaucoma examples of optic nerve rim losses and their evidence maps. The first to third rows are Example #1, Example #2, and Example #3. (a) fundus image. (b) label EM provided by teacher network. (c) EM generated by MTCLF. (d) visualized evidence map.

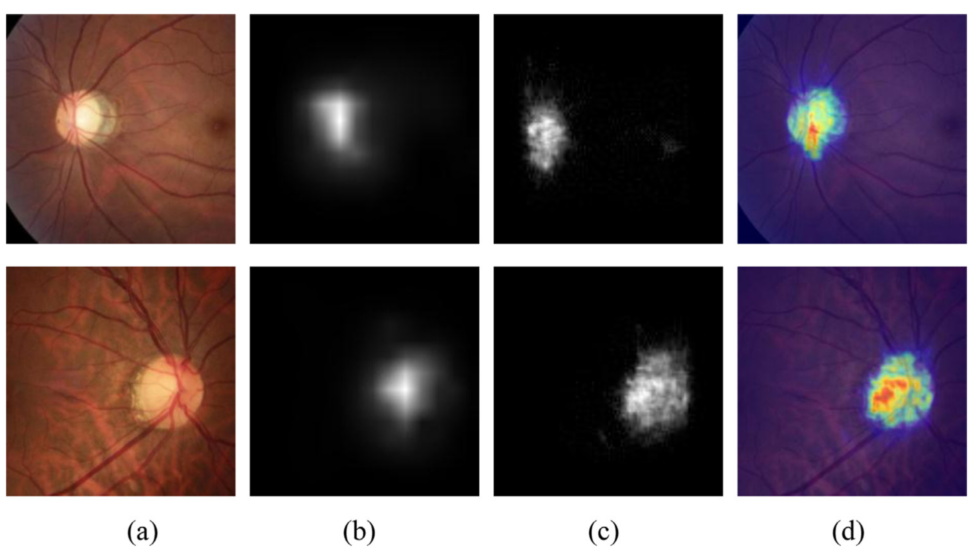

Fig. 10. Glaucoma examples of β-PPA and their evidence maps. The first row is Example #1, and the second row is Example #2. (a) fundus image. (b) label EM provided by teacher network. (c) EM generated by MTCLF. (d) visualized evidence map.

can capture this pathological change well, and its attention area is concentrated in the centre of the optic disc and the tangent areas above and below the optic nerve rim. In Example #2, the temporal optic nerve rim is severely lost, and the signs of tangency are obvious. These pathologically changed areas are consistent with the activation areas indicated by the evidence map. The image in Example #3 shows a severe loss of the optic nerve rim on the superior temporal side, and the evidence map accurately depicts this phenomenon.

β-PPA [6] is generally related to changes in the optic nerve due to glaucoma and can reflect the severity of glaucoma lesions. Fig. 10 shows some fundus images containing glaucoma with β-PPA. The key diagnostic areas reflected by the evidence map correspond to the β-PPA areas, which proves that β-PPA makes a key contribution to the diagnosis of glaucoma by MTCLF.

The evidence map reflects the basis for MTCLF to detect glaucoma, which is consistent with the diagnosis of clinical ophthalmologists paying attention to the morphology of the optic nerve rim, the trend of optic cup expansion, or β-PPA. Thus, visualizing the evidence map can help clinicians discover more undiscovered diagnostic areas from the images, such as β-PPA and optic nerve rim loss. In addition, evidence maps can actively assist doctors in understanding the changes in glaucoma tissue morphology and assist clinical glaucoma diagnoses. Overall, the evidence map output by MTCLF not only improves the performance of glaucoma diagnoses but also provides more detailed visualization results of the decision-making regions.

4.6. Discussion

From the experimental results based on the LAG, ORIGA, REFUGE and RIM-ONE-r2 datasets, it can be seen that the proposed MTCLF model achieves the best overall glaucoma screening performance in terms of the accuracy, sensitivity, specificity, AUC and F_2-score indicators. More importantly, MTCLF can maintain the balance between sensitivity and specificity while achieving the highest accuracy, which makes it better than the other latest algorithms, such as AG-CNN.

The key points that allow MTCLF to achieve excellent glaucoma screening performance are: i) the designed curriculum-based weighting coefficients; ii) the prior information and training guidance provided by the evidence map for the student network; iii) the designed unique teacher and student network architecture.

The combination of curriculum weighting coefficients θ and σ helps the student network focus on difficult training samples and misclassified samples with high similarity among classes and enhances its hard sample recognition and mining capabilities. In addition, the coefficients also play a role in balancing the distribution of training samples and adjusting the contribution of sample training. Therefore, the curriculum weighting coefficients significantly improve the sensitivity while maintaining stable specificity, which improves the performance of glaucoma screenings.

EM generated by the teacher network carries a prior information about the subtle lesions of glaucoma. The student network uses EM as labels for its evidence map prediction branch and uses EM as an attention map to reinforce the key related areas of its deep feature maps. Therefore, the glaucoma diagnostic performance of MTCLF is greatly improved.

The designed teacher network architecture uses the GC block self-attention module to reorganize the feature maps of different depths in ResNet. It not only helps to discover the evidence maps of key spatial features but also provides a guarantee for the generation of the curriculum. The dual-branch architecture and collaborative learning module of the student network ensure the accurate completion of multiple tasks. Therefore, the well-designed teacher-student network architecture can fully improve the glaucoma diagnosis performance of MTCLF under the joint training guidance of the designed curriculum weighting coefficients and evidence maps. Besides, MTCLF has good real-time performance. It takes 0.132s to infer a fundus image based on a personal computer operating environment with Intel Core i7-8750H CPU and GTX 1050TI GPU. Thus, MTCLF can effectively meet the requirements of large-scale clinical screening with its excellent glaucoma diagnostic performance and real-time performance.

5. Conclusions and Future Work

The MTCLF model proposed in this paper innovatively combines curriculum learning with a multitask CNN to achieve unbiased glaucoma diagnoses and predictions of evidence maps, assist clinical ophthalmologists in diagnosing glaucoma and provide model interpretability. MTCLF is a well-designed teacher-student framework. The trained teacher network is used to generate the label evidence map and the prior curriculum coefficient θ, and the student network is a multitask CNN that realizes the prediction of the evidence map and the unbiased diagnosis of glaucoma. The label evidence map serves as a bridge for information transmission between the teacher and student architecture, not only instructing the student network to generate evidence maps but also weighting the deep feature maps of the glaucoma prediction branches to provide sample prior knowledge. The designed a prior curriculum coefficient θ and self-feedback curriculum coefficient σ are used to dynamically adjust the loss function during the training of the student network so that the student network gradually focuses on hard samples, overcomes the influence of similar samples among classes, and balances the contributions of different samples to finally improve the performance of glaucoma diagnoses. The experimental results show that MTCLF's glaucoma diagnostic performance is superior to the state-of-the-art methods, it not only achieves the best performance in unbiased glaucoma diagnoses but also generates a reliable evidence map to help clinicians explore fine lesion areas.

In short, MTCLF provides a general framework that can be effectively applied to unbiased screening problems in other medical fields besides glaucoma screening. It can achieve the following positive effects:

i) MTCLF provides a new method for training models with unevenly distributed data, which can effectively alleviate the prediction bias caused by data imbalance.
ii) The design ideas of curriculum coefficients and the training strategy of the model are of great help in mining hard samples and improving the generalization ability of the model. Reducing FPR and FNR of predicting results is particularly important in the field of medical screening.
iii) MTCLF provides an effective solution for multi-task learning. It integrates the dual tasks of evidence map prediction and glaucoma screening in the student CNN network, and can achieve excellent results.
iv) MTCLF provides a new idea for visualizing the decision-making area of the CNN model. The activation map generated by the Grad CAM++ algorithm is used to assist the student network to learn to reconstruct the evidence map, so the evidence map output by the student network can provide more detailed decision-making area visualization results to better help clinicians explore fine lesion areas.

The current limitation is that the teacher-student architecture network is more dependent on the GPU. Future research on the knowledge distillation method for student networks is the key to get rid of the dependence on high-performance computing power and realize large-scale promotion.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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